

React Concepts Overview

Core Concepts:

1. React Basics – Library for building UI with components.
2. JSX – HTML-like syntax in JavaScript.
3. Components – Reusable UI building blocks.
4. Props – Data passed from parent to child.
5. State – Internal data that changes over time.
6. Event Handling – Handling user actions.
7. Conditional Rendering – Display UI based on conditions.
8. Lists and Keys – Rendering lists efficiently.

Intermediate Concepts:

9. Forms – Handling user input.
10. Controlled vs Uncontrolled – Who manages form data.
11. Lifting State Up – Sharing state between components.
12. Composition – Combining components.
13. Reconciliation – Efficient DOM updates.
14. Virtual DOM – Lightweight DOM copy.
15. Fragments – Group elements without extra nodes.

Advanced Concepts:

16. Hooks – `useState`, `useEffect`, etc.
17. Context API – Global state management.
18. HOC – Enhance components.
19. Render Props – Share logic via functions.
20. Portals – Render outside main DOM.
21. Error Boundaries – Handle errors gracefully.

Performance:

22. Memoization – Avoid re-renders.
23. Lazy Loading – Load when needed.
24. Code Splitting – Split app into chunks.

Ecosystem:

- 25. Routing – Navigation.
- 26. State Management – Redux, Zustand, Recoil.
- 27. API Integration – Fetching data.

Architecture:

- 28. SPA – Single page apps.
- 29. Lifecycle – Mount, Update, Unmount.
- 30. Separation of Concerns – Clean structure.

Testing & Tools:

- 31. Testing – Jest, React Testing Library.
- 32. Debugging – React DevTools.
- 33. Build Tools – Webpack, Vite.
- 34. Deployment – Netlify, Vercel.